

CS 456 – Mathematical Foundations of Machine Learning

Assignment 2 – Archisa Bhattacharya

Problem 1

Problem statement

Consider the following system of linear equations for some constants b and g :

$$1x + by + 0z = 1,$$

$$1x - 2y - 1z = 2,$$

$$0x + 1y + 1z = g.$$

- a) Which number b leads to a row exchange?
- b) Which number b makes the system singular (missing a third pivot)?
- c) For the b found in part (b), for which value of g does the system have infinitely many solutions?
- d) Find the solution of the system with b and g found in parts (b) and (c), if $z = 2$.

(a) Which number b leads to a row exchange?

(i) Write the augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & b & 0 & 1 \\ 1 & -2 & -1 & 2 \\ 0 & 1 & 1 & g \end{array} \right].$$

(ii) Eliminate x under the 1st pivot:

$$R_2 \leftarrow R_2 - R_1.$$

This gives

$$\left[\begin{array}{ccc|c} 1 & b & 0 & 1 \\ 0 & -2-b & -1 & 1 \\ 0 & 1 & 1 & g \end{array} \right].$$

(iii) The *normal* 2nd pivot is

$$-2 - b.$$

(iv) As long as it is not zero, it should be a pivot.

So a row exchange is necessary when

$$-2 - b = 0.$$

(v) Solve:

$$b = -2.$$

Therefore,

$$\boxed{b = -2}.$$

Extra note

This is only a *temporary* missing pivot. When $b = -2$, row 2 cannot pivot in column 2, but row 3 still has a 1 there, so we can swap rows.

(b) Which number b makes the system singular (missing a third pivot)?

(i) Start from

$$\left[\begin{array}{ccc|c} 1 & b & 0 & 1 \\ 0 & -2-b & -1 & 1 \\ 0 & 1 & 1 & g \end{array} \right].$$

(ii) We want to eliminate the 1 in row 3, column 2.

Use

$$R_3 \leftarrow (b+2)R_3 + R_2.$$

(iii) This gives

$$\left[\begin{array}{ccc|c} 1 & b & 0 & 1 \\ 0 & -2-b & -1 & 1 \\ 0 & 0 & b+1 & (b+2)g+1 \end{array} \right].$$

(iv) The possible 3rd pivot is

$$b+1.$$

(v) If $b+1=0$, there could be a missing 3rd pivot:

$$b+1=0.$$

(vi) Solve:

$$\boxed{b = -1}.$$

IMPORTANT NOTE

(i) Pivot temporarily disappears $\Rightarrow$ row swap may fix it.

(ii) Final pivot disappears $\Rightarrow A$ is singular.

(iii) The real question after that is: does the missing-pivot row become $0=0$ or $0=c \neq 0$?

(c) For the b found in part (b), for which value of g does the system have infinitely many solutions?

(i) From part (b),

$$b = -1.$$

(ii) Substitute $b = -1$ into the last row:

$$\left[\begin{array}{ccc|c} 0 & 0 & 0 & (1)g+1 \end{array} \right].$$

(iii) For infinitely many solutions, the missing-pivot row must become

$$0=0.$$

So we need

$$g+1=0.$$

(iv) Solve:

$$\boxed{g = -1}.$$

Extra note

Missing pivot:

$$0 = c, \quad c \neq 0 \quad \Rightarrow \quad \text{no solutions.}$$

Missing pivot:

$$0 = 0 \quad \Rightarrow \quad \text{free variable(s)} \quad \Rightarrow \quad \text{infinitely many solutions.}$$

(d) Find the solution with $b = -1$, $g = -1$, if $z = 2$.

(i) Substitute $b = -1$ and $g = -1$:

$$\begin{aligned}x - y &= 1, \\x - 2y - z &= 2, \\y + z &= -1.\end{aligned}$$

(ii) We are given

$$z = 2.$$

Use the third equation:

$$y + 2 = -1.$$

(iii) Solve for y :

$$y = -3.$$

(iv) Use the first equation:

$$x - y = 1.$$

Substitute $y = -3$:

$$x - (-3) = 1.$$

So

$$x + 3 = 1.$$

(v) Solve:

$$x = -2.$$

(vi) Therefore,

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2 \\ -3 \\ 2 \end{bmatrix}.$$

IMPORTANT NOTE

(i) Pivots tell us what kind of system we have.

(ii) Missing pivot + $0 = c$, $c \neq 0 \Rightarrow$ no solution.

(iii) Missing pivot + $0 = 0 \Rightarrow$ free variable(s) $\Rightarrow$ infinitely many solutions.

Problem 2

Problem statement

Find a basis for each of the four fundamental subspaces of

$$A = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ 0 & 0 & 2 & 4 \end{bmatrix}.$$

Step 1: RREF A

(i) Start with

$$A = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ 0 & 0 & 2 & 4 \end{bmatrix}.$$

(ii) Eliminate below the first pivot:

$$R_2 \leftarrow R_2 - 2R_1.$$

So

$$\begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 2 & 4 \end{bmatrix}.$$

(iii) Eliminate the duplicate third row:

$$R_3 \leftarrow R_3 - R_2.$$

So

$$\begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

(iv) Make the second pivot 1:

$$R_2 \leftarrow \frac{1}{2}R_2.$$

So

$$\begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

(v) Eliminate above the pivot in column 3:

$$R_1 \leftarrow R_1 - R_2.$$

Therefore,

$$\text{RREF}(A) = \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

(vi) The pivot columns are columns 1 and 3, so

$$\boxed{\text{rank}(A) = 2}.$$

(a) Space 1: The column space $C(A)$

(i) Look at the RREF and find the pivot columns:

$$\boxed{\text{pivot columns} = 1, 3}.$$

(ii) For the column space, go back to the *original* matrix A .

(iii) Take columns 1 and 3 of the original A :

$$\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix}.$$

(iv) Therefore,

$$\boxed{\mathcal{B}_{C(A)} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} \right\}.$$

(b) Space 2: The row space $C(A^T)$

(i) If we throw away the zero row, the nonzero rows of the RREF give a basis for the row space.

(ii) The nonzero rows are

$$\begin{bmatrix} 1 & 3 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & 1 & 2 \end{bmatrix}.$$

(iii) Therefore,

$$\boxed{\mathcal{B}_{C(A^T)} = \{ \begin{bmatrix} 1 & 3 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & 2 \end{bmatrix} \}.$$

(c) Space 3: The nullspace $N(A)$

(i) We need all vectors x satisfying

$$Ax = 0.$$

(ii) Use the RREF:

$$\begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

(iii) Read off the equations:

$$x_1 + 3x_2 = 0, \quad x_3 + 2x_4 = 0.$$

(iv) Solve for the pivot variables:

$$x_1 = -3x_2, \quad x_3 = -2x_4.$$

(v) x_2 and x_4 are free variables.

Let

$$x_2 = s, \quad x_4 = t.$$

(vi) Then

$$x = \begin{bmatrix} -3s \\ s \\ -2t \\ t \end{bmatrix}.$$

(vii) Split by s and t :

$$x = s \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}.$$

(viii) Therefore,

$$\mathcal{B}_{N(A)} = \left\{ \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}.$$

(d) **Space 4: Left nullspace $N(A^T)$**

(i) Since A has 3 rows, vectors in $N(A^T)$ have 3 entries:

$$y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}.$$

(ii) Since

$$\text{rank}(A) = 2,$$

and A has 3 rows,

$$\dim N(A^T) = 3 - 2 = 1.$$

So we are only looking for one basis vector.

(iii) Solve

$$A^T y = 0.$$

We have

$$A^T = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 6 & 0 \\ 1 & 4 & 2 \\ 2 & 8 & 4 \end{bmatrix}.$$

(iv) From the first equation,

$$y_1 + 2y_2 = 0.$$

So

$$y_1 = -2y_2.$$

(v) From the third equation,

$$y_1 + 4y_2 + 2y_3 = 0.$$

Substitute $y_1 = -2y_2$:

$$-2y_2 + 4y_2 + 2y_3 = 0.$$

So

$$2y_2 + 2y_3 = 0$$

and therefore

$$y_3 = -y_2.$$

(vi) Let

$$y_2 = t.$$

Then

$$y = \begin{bmatrix} -2t \\ t \\ -t \end{bmatrix} = t \begin{bmatrix} -2 \\ 1 \\ -1 \end{bmatrix}.$$

(vii) Therefore,

$$\mathcal{B}_{N(A^T)} = \left\{ \begin{bmatrix} -2 \\ 1 \\ -1 \end{bmatrix} \right\}.$$

Remember for exam

- (i) RREF first.
- (ii) Column space: find pivot columns in the RREF.
- (iii) Then take those same columns from the *original* A .
- (iv) Row space: use the nonzero rows of the RREF.
- (v) Nullspace: solve $Ax = 0$.
- (vi) Left nullspace: solve $A^T y = 0$.

A short version of the handwritten memory cue is:

Columns = originals, Rows = reduced, Nulls = solved.
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Problem 3

Problem statement

Suppose the complete solution to

$$Ax = \begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix}$$

is

$$x = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + c \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + d \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

- a) What is the dimension of the row space of A ?
- b) What is the matrix A ?
- c) Describe exactly all the vectors b for which $Ax = b$ can be solved. Do not just say that b must be in the column space of A .

(a) What is the dimension of the row space of A ?

(i) The complete solution is

$$x = x_p + x_n,$$

where

$$x_p = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$$

and

$$x_n = c \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + d \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

(ii) The two nullspace vectors

$$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

are independent.

Therefore,

$$\dim N(A) = 2.$$

(iii) x has 3 entries, so A has 3 columns.

(iv) Use rank-nullity:

$$\text{rank}(A) + \text{nullity}(A) = 3.$$

(v) Substitute nullity = 2:

$$\text{rank}(A) + 2 = 3.$$

Thus

$$\text{rank}(A) = 1.$$

(vi) The dimension of the row space equals the rank:

$$\dim C(A^T) = 1.$$

Extra note

The handwritten shortcut is:

$$\text{rank}(A) + \text{nullity}(A) = n.$$

Here:

$$\text{rank}(A) + 2 = 3.$$

(b) What is the matrix A ?

(i) Every row of A must be perpendicular to both nullspace vectors.

Let one row be

$$r = [r_1 \quad r_2 \quad r_3].$$

(ii) Use the first nullspace vector:

$$r \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = 0.$$

So

$$r_1 + r_2 = 0.$$

Therefore,

$$r_2 = -r_1.$$

(iii) Use the second nullspace vector:

$$r \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = 0.$$

So

$$r_3 = 0.$$

(iv) Therefore every row of A has the form

$$r = \alpha [1 \quad -1 \quad 0].$$

(v) Since

$$Ax_p = \begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix}, \quad x_p = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix},$$

use the particular solution to determine the row multiples.

(vi) If row i is

$$\alpha_i [1 \quad -1 \quad 0],$$

then

$$\alpha_i [1 \quad -1 \quad 0] \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} = 2\alpha_i.$$

(vii) Match this with

$$\begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix}.$$

So

$$2\alpha_1 = 2, \quad 2\alpha_2 = 4, \quad 2\alpha_3 = 4.$$

Thus

$$\alpha_1 = 1, \quad \alpha_2 = 2, \quad \alpha_3 = 2.$$

(viii) Therefore,

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 2 & -2 & 0 \\ 2 & -2 & 0 \end{bmatrix}.$$

(c) Describe exactly all vectors b for which $Ax = b$ can be solved.

(i) Use the matrix from part (b):

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 2 & -2 & 0 \\ 2 & -2 & 0 \end{bmatrix}.$$

(ii) Its columns are

$$\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \quad \begin{bmatrix} -1 \\ -2 \\ -2 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

(iii) Every column is a multiple of

$$\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

So every vector that can appear as Ax must lie on that line.

(iv) Therefore $Ax = b$ is solvable exactly when

$$b = t \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

for some $t \in \mathbb{R}$.

(v) In entry form,

$$b = \begin{bmatrix} t \\ 2t \\ 2t \end{bmatrix}, \quad t \in \mathbb{R}.$$

(vi) Equivalently,

$$b_2 = 2b_1, \quad b_3 = 2b_1.$$

IMPORTANT NOTE

The question says *do not just say* “ b is in the column space of A .”
So the final answer should explicitly describe the vectors:

$$b = (t, 2t, 2t)^T.$$

Problem 4

Problem statement

a) Find an orthonormal basis for the subspace S of $\mathbb{R}^4$ spanned by

$$a_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \quad a_2 = \begin{bmatrix} 3 \\ 1 \\ 1 \\ 3 \end{bmatrix}, \quad a_3 = a_1 + a_2.$$

b) Find the closest vector p in that subspace S to

$$b = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

(a) Find an orthonormal basis for S .

(i) Notice that

$$a_3 = a_1 + a_2.$$

So a_3 does not add a new independent direction.

Therefore,

$$S = \text{span}\{a_1, a_2\}.$$

(ii) Start Gram–Schmidt with

$$u_1 = a_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

(iii) Find its norm:

$$\|u_1\| = \sqrt{1^2 + 1^2 + 1^2 + 1^2} = 2.$$

(iv) Normalize:

$$q_1 = \frac{u_1}{\|u_1\|} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

(v) Now remove from a_2 the part that points in the q_1 direction:

$$u_2 = a_2 - (q_1^T a_2)q_1.$$

(vi) Compute the coefficient:

$$q_1^T a_2 = \frac{1}{2}(3 + 1 + 1 + 3) = 4.$$

(vii) Therefore,

$$(q_1^T a_2)q_1 = 4q_1 = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix}.$$

(viii) Subtract:

$$u_2 = \begin{bmatrix} 3 \\ 1 \\ 1 \\ 3 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}.$$

(ix) Find its norm:

$$\|u_2\| = \sqrt{1 + 1 + 1 + 1} = 2.$$

(x) Normalize:

$$q_2 = \frac{u_2}{\|u_2\|} = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}.$$

(xi) Therefore an orthonormal basis is

$$\mathcal{B}_S = \left\{ \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

Extra note

There is no third orthonormal basis vector because

$$a_3 = a_1 + a_2.$$

It is already in the span of the first two vectors.

(b) Find the closest vector $p \in S$ to b .

(i) “Closest vector in the subspace” means the orthogonal projection of b onto S .

(ii) Because $\{q_1, q_2\}$ is an orthonormal basis,

$$p = (b^T q_1)q_1 + (b^T q_2)q_2.$$

(iii) Use

$$b = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Then

$$b^T q_1 = \frac{1}{2}.$$

(iv) Also,

$$b^T q_2 = \frac{1}{2}.$$

(v) Therefore,

$$p = \frac{1}{2}q_1 + \frac{1}{2}q_2.$$

(vi) Substitute q_1 and q_2 :

$$p = \frac{1}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \frac{1}{4} \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}.$$

(vii) Add:

$$p = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \end{bmatrix}.$$

Therefore,

$$p = \begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \end{bmatrix}.$$

Remember for exam

- (i) Gram-Schmidt makes the spanning vectors orthogonal.
- (ii) Normalize each nonzero orthogonal vector to make the basis orthonormal.
- (iii) Once the basis is orthonormal, projection is easy:

$$\text{proj}_S(b) = \sum_i (b^T q_i) q_i.$$

(iv) So:

$$\text{closest vector in } S = \text{projection onto } S.$$